Anisotropic Surface-Mapped Coarse-Grained Beads for Molecular Systems

VSEPR-SIM Project

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Abstract

Traditional coarse-grained (CG) models frequently reduce molecular fragments to isotropic point particles, preserving total mass and charge while discarding directional structure. This work proposes an intermediate representation: anisotropic surface-mapped beads, in which grouped atoms are projected to a single center while retaining directional interaction information via a surface descriptor. Using benzene as a canonical example, we demonstrate how this approach preserves planar symmetry and directional response while maintaining computational efficiency. The method provides a bridge between atomistic fidelity and reduced-order modeling suitable for multi-scale simulation.

1 Introduction

Coarse-grained modeling aims to reduce computational complexity by grouping atoms into larger interaction sites. Standard approaches represent each group as an isotropic particle with effective interaction potentials:

$$U_{ij} = U(r_{ij}) \tag{1}$$

While efficient, this approximation discards directional information inherent to many molecular structures. For planar or anisotropic systems such as benzene, this leads to loss of physically relevant behavior.

This work introduces a surface-mapped bead representation, preserving directional characteristics while retaining a single-site description.

2 Atomistic to Coarse-Grained Mapping

Let a bead B be formed from atoms $i \in S_B$.

The center of mass is defined as:

$$\mathbf{R}_B = \frac{1}{M_B} \sum_{i \in S_B} m_i \mathbf{r}_i \tag{2}$$

with total mass:

$$M_B = \sum_{i \in S_B} m_i \tag{3}$$

and total charge:

$$Q_B = \sum_{i \in S_B} q_i \tag{4}$$

A local coordinate frame is constructed using the inertia tensor of the atom group.

3 Surface Descriptor Representation

Instead of representing the bead as isotropic, we define a directional surface descriptor on a spherical shell:

$$S_B(\theta, \phi) \tag{5}$$

where (θ, ϕ) are angular coordinates in the bead's local frame.
This function encodes anisotropic properties such as:

- effective density
- interaction strength
- electrostatic response

A practical formulation is obtained by sampling interaction with a probe particle:

$$U_B(r, \theta, \phi) = \sum_{i \in S_B} U_{i,p}(|\mathbf{x} - \mathbf{r}_i|) \tag{6}$$

where $\mathbf{x} = \mathbf{R}_B + r\hat{n}(\theta, \phi)$.

4 Compressed Representation

To reduce storage cost, the surface descriptor may be expanded in spherical harmonics:

$$S_B(\theta, \phi) = \sum_{\ell, m} c_{\ell m} Y_{\ell m}(\theta, \phi) \tag{7}$$

Low-order expansions are sufficient for many rigid molecular fragments.

5 Application to Benzene

Benzene is a planar molecule with strong anisotropy:

- circular in-plane symmetry
- distinct out-of-plane behavior
- non-uniform interaction profile

A surface-mapped bead preserves:

- planar geometry
- orientation-dependent interactions
- effective size and shape

while reducing the system from 12 atoms to a single interaction site.

6 Interaction Model

Interactions between beads are defined using directional dependence:

$$U = U(r, \Omega_B, \Omega_C) \tag{8}$$

or via surface coupling:

$$U \approx \int S_B(\theta, \phi) S_C(\theta', \phi') K(r)\, d\Omega \tag{9}$$

where $K(r)$ is a radial kernel.

7 Dynamic Consistency and Updates

The surface descriptor remains valid under small deformations. However, recomputation is required when:

- bonds are broken or formed
- large conformational changes occur
- charge distribution changes significantly

8 Advantages and Limitations

8.1 Advantages

- retains directional physics
- reduces atom count significantly
- enables efficient reuse of precomputed structure

8.2 Limitations

- state dependence of effective interactions
- additional preprocessing cost
- need for descriptor compression

9 Orientation-Coupled Interaction Framework

The surface descriptor $S_B(\theta, \phi)$ encodes the intrinsic directional structure of an individual coarse-grained bead. However, this representation alone is insufficient to define interactions between beads, since intermolecular behavior depends not only on internal anisotropy but also on the *relative orientation* between interacting units.

To construct a physically meaningful coarse-grained model, the interaction framework must therefore couple radial separation with orientational degrees of freedom, enabling the emergence of direction-dependent energetics such as stacking, alignment, and angular selectivity.

9.1 Bead Orientation Representation

Each bead B, formed from an underlying atomic subset, is endowed with a local orthonormal frame $Q_B = \{\hat{e}_1, \hat{e}_2, \hat{e}_3\}$ derived from the principal axes of its inertia tensor. This construction ensures that the orientation is both physically grounded and invariant under rigid-body transformations of the atomistic configuration.

For planar or near-planar structures, such as benzene, the orientation can be reduced to a primary unit vector $\hat{n}_B$, defined as the normal to the molecular plane. This provides a minimal yet sufficient representation of anisotropy for systems dominated by planar symmetry.

Given two interacting beads A and B, define the relative displacement vector:

$$\mathbf{r} = \mathbf{R}_B - \mathbf{R}_A, \quad r = |\mathbf{r}|, \quad \hat{r} = \frac{\mathbf{r}}{r} \tag{10}$$

The orientational relationship between the beads is captured through the scalar invariants:

$$\cos\theta_A = \hat{n}_A \cdot \hat{r}, \quad \cos\theta_B = \hat{n}_B \cdot \hat{r}, \quad \cos\phi = \hat{n}_A \cdot \hat{n}_B \tag{11}$$

These quantities form a reduced set of rotationally invariant descriptors that fully characterize the relative alignment between two axisymmetric anisotropic beads.

9.2 Reduced Interaction Model

A tractable first-order anisotropic interaction potential may be constructed by augmenting a baseline radial interaction with orientation-dependent corrections:

$$U_{\text{eff}}(r, \theta_A, \theta_B, \phi) = U_0(r) + \lambda_1 f_1(r)\cos\phi + \lambda_2 f_2(r)\cos^2\theta_A + \lambda_3 f_3(r)\cos^2\theta_B \tag{12}$$

Here, $U_0(r)$ represents an isotropic reference potential (e.g., Lennard-Jones or Morse), while the functions $f_i(r)$ modulate the strength of angular contributions as a function of separation. The coefficients λ_i control the relative importance of different orientational modes.

This formulation introduces anisotropic sensitivity while preserving a low-dimensional parameter space, enabling efficient evaluation in large-scale simulations. Importantly, the chosen angular terms correspond to physically interpretable alignment tendencies, such as parallel orientation ($\cos\phi$) and alignment with the intermolecular axis ($\cos\theta$ terms).

9.3 Surface Descriptor Coupling

While the reduced model provides computational efficiency, higher-fidelity interactions may be derived directly from the surface descriptors themselves. In this formulation, the interaction energy arises from a convolution-like coupling between directional surface fields:

$$U \approx \int S_A(\theta, \phi)\, S_B(\theta', \phi')\, K(r)\, d\Omega \tag{13}$$

where $K(r)$ is a radial kernel governing distance dependence, and the angular coordinates (θ', ϕ') are obtained by transforming bead B's local frame into that of bead A.

This representation effectively treats each bead as a distributed interaction surface rather than a point object, allowing fine angular structure—including multipolar features and localized interaction regions—to contribute directly to the interaction energy.

In practice, this integral may be evaluated using truncated spherical harmonic expansions, reducing the interaction to a finite set of mode couplings.

9.4 Torque and Rotational Dynamics

Because the interaction potential depends explicitly on orientation, translational forces alone do not fully describe system dynamics. Rotational degrees of freedom must be incorporated to ensure physical consistency.

The torque acting on bead B is given by:

$$\boldsymbol{\tau}_B = -\frac{\partial U}{\partial \Omega_B} \tag{14}$$

where Ω_B denotes the rotational configuration of the bead, parameterized for example by Euler angles or quaternions.

This torque induces rotational motion according to rigid-body dynamics, enabling phenomena such as alignment, libration, and orientational relaxation to emerge naturally from the interaction model.

9.5 Application to Benzene Dimer Configurations

The orientation-dependent framework is particularly relevant for systems such as benzene dimers, where interaction energy is strongly modulated by relative alignment.

The model captures several well-known configurations:

- **Face-to-face stacking:** $\cos\phi \approx 1$, $\theta_A \approx \theta_B \approx 0$
- **Edge-to-face (T-shaped):** $\theta_A \approx \pi/2$, $\theta_B \approx 0$
- **Offset stacking:** intermediate angular values with lateral displacement

These configurations correspond to distinct energetic minima arising from π-π interactions and electrostatic anisotropy. Such behavior cannot be reproduced within isotropic coarse-grained models, highlighting the necessity of orientation-aware formulations.

9.6 Model Hierarchy

The proposed framework supports a systematic hierarchy of interaction models with increasing fidelity:

- **Isotropic:** $U(r)$
- **Axisymmetric anisotropic:** $U(r, \theta_A, \theta_B, \phi)$
- **Descriptor-resolved:** $U(r, \Omega_A, \Omega_B, S_A, S_B)$

This hierarchy enables adaptive resolution strategies, where simpler models are used for large-scale exploration and more detailed representations are invoked when directional effects become dominant.

In this sense, the orientation-coupled interaction framework serves as a bridge between classical coarse-grained potentials and fully resolved anisotropic interaction fields.

9.7 Descriptor Enrichment for Complex Anisotropic Structures

The spherical harmonic representation

$$S_B(\theta, \phi) = \sum_{\ell,m} c_{\ell m} Y_{\ell m}(\theta, \phi) \tag{15}$$

provides a compact and efficient description of moderately anisotropic systems. However, for highly structured and sterically complex entities—such as coordination clusters with bulky ligands—low-order expansions are insufficient. In these cases, the angular variation of interaction properties is sharp, non-uniform, and often dominated by localized features.

9.7.1 Higher-Order Angular Resolution

To accurately capture directional structure, the spherical harmonic expansion must be extended to higher orders:

$$S_B(\theta, \phi) = \sum_{\ell=0}^{\ell_{\max}} \sum_{m=-\ell}^{\ell} c_{\ell m} Y_{\ell m}(\theta, \phi) \tag{16}$$

where $\ell_{\max}$ is chosen based on the angular complexity of the system. Larger values of $\ell_{\max}$ allow representation of sharp steric occlusion and narrow interaction channels, at the cost of increased computational overhead.

9.7.2 Multi-Channel Surface Representation

A single scalar descriptor is generally insufficient to encode the diverse physical interactions present in complex systems. Instead, the surface descriptor is extended to a vector-valued representation:

$$S_B(\theta, \phi) \to \left\{ S_B^{(\text{steric})}(\theta, \phi),\ S_B^{(\text{elec})}(\theta, \phi),\ S_B^{(\text{disp})}(\theta, \phi) \right\} \tag{17}$$

Each channel captures a distinct physical contribution:

- $S^{(\text{steric})}$: geometric accessibility and excluded volume
- $S^{(\text{elec})}$: electrostatic potential or charge distribution
- $S^{(\text{disp})}$: dispersion and non-bonded interaction strength

This decomposition enables independent modeling and weighting of interaction mechanisms, improving both interpretability and accuracy.

9.7.3 Full Orientation Representation

For systems with three-dimensional anisotropy, a single orientation vector is insufficient. Instead, each bead is associated with a full orthonormal frame:

$$Q_B = \{\hat{e}_1, \hat{e}_2, \hat{e}_3\} \tag{18}$$

derived from the principal axes of the inertia tensor. This allows the surface descriptor to be consistently defined in the bead's local coordinate system and properly transformed during interaction evaluation.

9.7.4 Implications for Interaction Modeling

The enriched descriptor directly affects the interaction formulation. The effective interaction between two beads becomes:

$$U \approx \int \sum_k S_A^{(k)}(\theta, \phi)\, S_B^{(k)}(\theta', \phi')\, K_k(r)\, d\Omega \tag{19}$$

where k indexes interaction channels and $K_k(r)$ are channel-specific radial kernels.
This formulation enables:

- directional steric screening
- anisotropic electrostatic interactions
- orientation-dependent dispersion

9.7.5 Modeling Considerations

The introduction of higher-order and multi-channel descriptors increases both representational power and computational cost. Practical implementations must therefore balance:

- angular resolution ($\ell_{\max}$)
- number of descriptor channels
- evaluation efficiency

Adaptive schemes, in which descriptor complexity is increased only for highly anisotropic systems, may provide an effective compromise.

9.7.6 Physical Interpretation

At this level of representation, the bead is no longer a point particle with an effective potential, but rather a compressed representation of a three-dimensional interaction field. The modeling objective is therefore not simply reduction, but controlled preservation of physically relevant directional structure.

10 Model Selection Guidelines

To balance computational efficiency and physical fidelity, the interaction model is applied using a two-tier hierarchical strategy based on structural regime rather than a fixed atom-count threshold.

10.1 Tier 1: Reduced Anisotropic Model

The reduced orientation-coupled model is used for systems with low angular complexity. This includes structures that are:

- rigid and compact
- smoothly anisotropic
- well-described by a single dominant axis or simple frame

- not dominated by localized interaction sites

Typical examples include aromatic ring systems such as benzene, naphthalene, pyridine, and phenol.

The interaction is represented using a low-dimensional orientation-dependent potential:

$$U_{\text{eff}}(r, \theta_A, \theta_B, \phi) \tag{20}$$

This model is appropriate when:

- low-order spherical harmonic expansion is sufficient
- a small number of angular invariants captures the interaction behavior
- computational efficiency is prioritized

10.2 Tier 2: Enriched Descriptor Model

The enriched descriptor-based model is used for systems with high angular complexity and heterogeneous interaction surfaces. This includes structures that are:

- strongly three-dimensional
- sterically shielded or directionally gated
- chemically heterogeneous
- metal-centered or coordination-driven
- characterized by non-uniform surface accessibility

Typical examples include organometallic clusters, coordination complexes, and bulky substituted systems.

The bead is represented using a multi-channel surface descriptor:

$$S_B(\theta, \phi) \to \left\{ S^{(\text{steric})},\, S^{(\text{elec})},\, S^{(\text{disp})} \right\} \tag{21}$$

with a full orientation frame:

$$Q_B = \{\hat{e}_1, \hat{e}_2, \hat{e}_3\} \tag{22}$$

This model is required when:

- interaction depends on angular accessibility
- low-order angular representations produce large error
- multiple physical interaction modes must be resolved

10.3 Selection Heuristic

As a practical guideline, the enriched model is typically activated for systems with:

- atom count $N \gtrsim 18$
- presence of a metal center

However, the definitive criterion is the angular complexity of the surface descriptor. Systems that are poorly approximated by low-order angular representations must be promoted to the enriched model regardless of size or composition.

10.4 Summary

The reduced model provides efficiency for structurally simple systems, while the enriched model preserves directional physics in complex environments. This hierarchical approach enables adaptive resolution across a wide range of molecular and materials systems.

11 Unified Descriptor Strategy

The preceding model selection guidelines introduced a two-tier hierarchical strategy, distinguishing between a reduced orientation-coupled model and an enriched multi-channel descriptor model. While conceptually useful, this formulation relies on a discrete architectural switch—triggered by atom count, metal center presence, or angular complexity—that introduces a discontinuous boundary between model regimes.

This section presents a refinement: rather than maintaining two separate data models with distinct interaction machinery, all systems are represented within a *single descriptor-based framework*. Model complexity is adjusted through controlled truncation rather than by changing the underlying formalism. The result is a continuous, unified architecture in which the previous "tiers" emerge as natural truncation levels of one model family.

11.1 Motivation

Hard switching between separate coarse-grained interaction models introduces several undesirable properties:

- **Discontinuity:** The energy surface is non-smooth at the model boundary, complicating optimization and dynamics.
- **Non-transferability:** Parameters calibrated in one model regime do not apply in the other.
- **Error opacity:** It is difficult to compare model quality across regimes when the representations differ.
- **Arbitrary thresholds:** Heuristic cutoffs (e.g., $N \geq 18$, metal center present) do not correspond to a smooth physical variation.

A unified formalism avoids these issues by ensuring that model enrichment occurs within a single continuous framework.

11.2 Unified Formalism

All beads, regardless of structural complexity, are represented using the same multi-channel spherical harmonic descriptor:

$$S_B^{(k)}(\theta, \phi) = \sum_{\ell=0}^{\ell_{\max}} \sum_{m=-\ell}^{\ell} c_{\ell m}^{(k)} Y_{\ell m}(\theta, \phi) \tag{23}$$

where k indexes the interaction channel (steric, electrostatic, dispersion) and $\ell_{\max}$ controls angular resolution.

Each bead carries a full orientation frame $Q_B = \{\hat{e}_1, \hat{e}_2, \hat{e}_3\}$ and a set of descriptor coefficients $\{c_{\ell m}^{(k)}\}$ for each active channel.

The interaction energy between two beads is defined by a single universal form:

$$U = F\Big(r,\, Q_A,\, Q_B,\, \{c_{\ell m}^{(k)}\}\Big) \tag{24}$$

which evaluates as:

$$U = U_0(r) + \sum_k \lambda_k \left(\sum_{\ell,m} c_{\ell m}^{(k)}(A)\, c_{\ell m}^{(k)}(B) \right) K_k(r) \tag{25}$$

where $U_0(r)$ is the isotropic baseline, $K_k(r)$ are channel-specific radial kernels, and λ_k are coupling weights.

This formulation subsumes both the reduced and enriched models as special cases:

- **Reduced regime:** small $\ell_{\max}$, single active channel, few nonzero coefficients
- **Enriched regime:** larger $\ell_{\max}$, multiple active channels, richer coefficient structure

The transition between these regimes is not a model switch, but a change in the number of active coefficients within the same mathematical object.

11.3 Adaptive Truncation

Rather than selecting a model tier via fixed heuristic thresholds, the unified framework determines the appropriate resolution level through *descriptor residual analysis*.

Given a bead's atomistic structure, the procedure is:

1. Fit a low-order descriptor ($\ell_{\max} = \ell_0$)
2. Compute the reconstruction residual $\varepsilon(\ell_0)$
3. If $\varepsilon > \varepsilon_{\text{tol}}$, increase $\ell_{\max}$ and activate additional channels
4. Repeat until residual is below tolerance or maximum resolution is reached

The reconstruction residual is defined as:

$$\varepsilon(\ell_{\max}) = \frac{\sum_i |S_{\text{ref}}(\hat{n}_i) - S_{\ell_{\max}}(\hat{n}_i)|^2}{\sum_i |S_{\text{ref}}(\hat{n}_i)|^2} \tag{26}$$

where S_{ref} is the probe-sampled reference surface and $S_{\ell_{\max}}$ is the truncated spherical harmonic reconstruction.

This approach ensures that complexity is added continuously in response to representational need, rather than through fixed structural cutoffs.

11.4 Resolution Levels

The unified descriptor supports a continuous hierarchy of resolution levels, which replaces the discrete tier concept:

- **Level 0 (Isotropic):** Only c_{00} retained; equivalent to a point particle with effective radius
- **Level 1 (Axial):** $\ell_{\max} \leq 2$, single channel; captures dominant planar or axial anisotropy
- **Level 2 (Moderate):** $\ell_{\max} \leq 4$, one or two channels; sufficient for rigid aromatic systems
- **Level 3 (Enriched):** $\ell_{\max} \leq 8$, all channels active; required for sterically complex or metal-centered systems

This hierarchy is not a set of discrete models but a continuum of truncation depths within one mathematical framework.

11.5 Advantages over Discrete Tier Selection

The unified descriptor strategy provides several advantages:

- **Smooth energy surface:** No discontinuities at model boundaries
- **Single optimization space:** Parameters, coefficients, and errors are all defined in the same representation
- **Measurable error:** Reconstruction residual provides a quantitative basis for resolution selection
- **Hierarchical refinement:** Start coarse, refine only where needed
- **Transferability:** The same descriptor object and energy machinery apply across all chemical systems

11.6 Relationship to Previous Tier Concept

The two-tier model selection guidelines presented in the preceding section remain valid as a practical engineering heuristic. However, they are now understood as approximate indicators of where a system falls within the continuous resolution hierarchy, rather than as triggers for architectural switching.

In implementation, the "reduced" regime corresponds to a low-order truncation of the unified descriptor, and the "enriched" regime corresponds to a higher-order truncation with additional active channels. The underlying data model and energy machinery are identical in both cases.

12 Atomistic Preparation Layer

The unified descriptor strategy establishes a mathematically coherent coarse-grained formalism. However, the quality of any coarse-grained representation is bounded by the quality of the atomistic input from which it is derived. If the atomistic preparation is unreliable—with incomplete topology, missing charges, or ill-defined local frames—the descriptor machinery inherits this uncertainty and the resulting coarse-grained model is untrustworthy regardless of its mathematical sophistication.

This section formalizes the atomistic preparation layer as a necessary prerequisite for reliable descriptor generation.

12.1 Purpose

The atomistic namespace exists to answer a fundamentally different question than the coarse-grained namespace:

- **Atomistic:** "What is this structure, exactly?"
- **Coarse-grained:** "How do I compress this structure into a transferable reduced object?"

The atomistic layer is the source-of-truth layer that owns:

- atoms, bonds, and formal charges
- local geometry and coordination environment
- ring detection and aromaticity flags
- molecular graph and canonical fragment decomposition
- local frames derived from real structure

Its outputs are consumed by the coarse-grained layer, but the reverse dependency must never exist. The atomistic layer must not depend on or import any coarse-grained types.

12.2 Scale Boundary: The Fragment View

Rather than allowing the coarse-grained layer to rummage through raw atom lists, a clean bridge object is defined at the scale boundary. This object—the *fragment view*—packages all atomistic information needed for descriptor construction into a single, validated interface.

The fragment view contains:

- atom records: position, atomic number, charge, flags
- bond records: atom indices and bond order
- aggregate properties: total mass, total charge
- a local coordinate frame derived from the inertia tensor
- structural classification flags: aromatic, cyclic, organometallic

The coarse-grained layer receives a fragment view and constructs a bead and its unified descriptor from it. This enforces a one-directional data flow:

$$\text{atomistic structure} \xrightarrow{\text{fragment view}} \text{unified descriptor} \rightarrow \text{bead interactions} \tag{27}$$

12.3 Validation and Status Codes

The atomistic preparation layer must produce either a valid fragment view or an explicit failure with a reason code. Silent propagation of invalid inputs is prohibited.

Failure modes include:

- invalid geometry (degenerate or collapsed atom positions)
- missing charges (incomplete charge assignment)
- ill-defined frame (insufficient atom count or degeneracy)
- unsupported topology (unrecognized bonding pattern)

Each failure mode produces a specific status code, making debugging tractable and preventing the coarse-grained layer from inheriting garbage.

12.4 Architectural Implications

The introduction of a formal scale boundary has several consequences:

- **Dependency asymmetry:** The atomistic namespace never imports from `coarse_grain`. The coarse-grained namespace depends on atomistic bridge types.
- **Testability:** The fragment view can be independently validated before descriptor construction.
- **Reusability:** The atomistic preparation layer supports additional consumers beyond coarse-graining: reference calculations, backmapping, validation against exact structure, fragment classification, and hybrid atomistic/CG regions.
- **Error traceability:** If a descriptor is incorrect, the fragment view provides the exact atomistic input that produced it.

12.5 Relationship to Descriptor Generation

The unified descriptor is constructed from a fragment view through the following chain:

1. Atomistic layer produces a validated `FragmentView`
2. The fragment view provides positions, charges, and frame to the mapping layer
3. The mapping layer generates probe-based surface fields
4. Surface fields are projected onto spherical harmonics to produce descriptor coefficients
5. The reconstruction residual determines whether the current truncation is sufficient

This chain ensures that every descriptor coefficient is traceable back through the fragment view to specific atoms in the original structure. The anti-black-box contract is maintained across the scale boundary.

13 Anisotropic Bead Model — Implementation Specification

This section formalises the unified coarse-grained descriptor framework with adaptive spherical harmonic resolution. All bead instances conform to a single, rigidly defined state structure; no dual-model branching is permitted.

13.1 Core Data Structure

A bead B is defined by the canonical tuple

$$B = \{\mathbf{R}_B,\ \mathbf{Q}_B,\ \{c_{\ell m}^{(k)}\},\ M_B,\ Q_B^{\text{tot}}\} \tag{28}$$

where $\mathbf{R}_B$ is the centre of mass, $\mathbf{Q}_B$ is an orthonormal 3×3 orientation frame, $\{c_{\ell m}^{(k)}\}$ are per-channel real spherical harmonic coefficients, M_B is the total mass, and Q_B^{tot} is the total charge.

Three physical channels are defined:

$$\begin{array}{ll} k=0 & \text{Steric (excluded-volume / atomic density)} \\ k=1 & \text{Electrostatic (Coulomb potential at surface)} \\ k=2 & \text{Dispersion (Lennard-Jones field at surface)} \end{array}$$

Invariants:

- $\mathbf{Q}_B$ must remain orthonormal; enforce via Gram–Schmidt at every mutation site.
- $\ell_{\max}$ must be consistent per channel within a given bead.
- Memory layout must be contiguous to enable SIMD vectorisation in downstream kernels.

13.2 Descriptor Generation Pipeline

The pipeline maps a fully specified atomistic fragment to a fully populated bead. All steps are deterministic; stochastic initialisation is not permitted.

1. **Centre computation:** $\mathbf{R}_B = M_B^{-1}\sum_{i\in S_B} m_i\mathbf{r}_i$
2. **Local frame:** $\mathbf{Q}_B$ via inertia tensor diagonalisation (Jacobi)
3. **Channel projection:** For each channel k, sample the surface field and project onto real spherical harmonics: $c_{\ell m}^{(k)} = \int S^{(k)}(\hat{n})\, Y_{\ell m}(\hat{n})\, d\Omega$
4. **Scalar aggregation:** $M_B = \sum m_i, \quad Q_B^{\text{tot}} = \sum q_i$

13.3 Channel Sampling Definitions

Channel	Physical Definition
Steric	Excluded-volume / atomic density projected onto bead surface
Electrostatic	Coulomb potential evaluated at surface points from partial charges
Dispersion	Lennard-Jones field contribution at bead surface

13.4 Interaction Engine

The pairwise interaction energy between beads A and B is evaluated in harmonic space. Runtime surface integration is not employed.

$$U_{AB}(r, \mathbf{Q}_A, \mathbf{Q}_B) = \sum_k I_{AB}^{(k)}(r, \mathbf{Q}_A, \mathbf{Q}_B) \tag{29}$$

The per-channel interaction requires rotating the coefficients of B into the local frame of A via Wigner-D matrices for real spherical harmonics, then summing:

$$I_{AB}^{(k)} = \sum_{\ell=0}^{\ell_{\max}} \sum_{m=-\ell}^{\ell} c_{\ell m}^{(k),A} \, \tilde{c}_{\ell m}^{(k),B} \, K_k(\ell, r) \tag{30}$$

where $\tilde{c}_{\ell m}^{(k),B}$ denotes the rotated coefficients and $K_k(\ell, r)$ is the per-channel, per-ℓ radial kernel:

$$K_{\text{steric}}(\ell, r) = \frac{e^{-\alpha_s r}}{1+\ell} \tag{31}$$

$$K_{\text{electrostatic}}(\ell, r) = \frac{1}{r^{\ell+1}} \tag{32}$$

$$K_{\text{dispersion}}(\ell, r) = \frac{-C_6}{r^{6+\ell}} \tag{33}$$

13.5 Adaptive Complexity

Resolution is adjusted per-bead per-channel by measuring reconstruction fidelity and incrementing $\ell_{\max}$ when the error metric exceeds a threshold. This eliminates dual-model architectures; a single model instance spans the full complexity range.

$$\varepsilon_B^{(k)} = \frac{\sum_i \left| S_{\text{ref}}^{(k)}(\hat{n}_i) - \tilde{S}^{(k)}(\hat{n}_i) \right|^2}{\sum_i \left| S_{\text{ref}}^{(k)}(\hat{n}_i) \right|^2} \tag{34}$$

If $\varepsilon_B^{(k)} > \varepsilon_{\text{tol}}$, then $\ell_{\max}^{(k)} \leftarrow \ell_{\max}^{(k)} + \Delta\ell$ and the channel is reprojected.
Expected behaviour:

- Small or symmetric molecules converge at low $\ell_{\max}$.
- Large or anisotropic structures self-select higher harmonic resolution.
- No branching into separate model classes—only resolution scaling.

13.6 System-Level Integration

The components specified above constitute a self-consistent physical model. Compatible integration targets include:

- MD integrators (velocity Verlet, Langevin)
- Monte Carlo sampling (Metropolis–Hastings)
- Bulk system simulations at arbitrary N

Required next steps:

1. Precompute Wigner-D rotation operators; cache per angular momentum order
2. Implement neighbour lists to reduce interaction evaluation from $O(N^2)$ to $O(N)$
3. Validate against benzene π-stacking reference energies
4. Validate against naphthalene crystal packing geometry
5. Validate against organometallic coordination geometries

14 Formal Bead State Decomposition

The canonical bead state

$$B = \{\mathbf{R}_B,\ \mathbf{Q}_B,\ \{c_{\ell m}^{(k)}\},\ M_B,\ Q_B^{\text{tot}}\} \tag{35}$$

is not merely a tuple of numbers. Each variable occupies a distinct physical role, obeys a distinct update rule, and serves a distinct architectural purpose. This section decomposes the state into its constituent layers and establishes the ownership, mutability, and numerical responsibilities of every term.

14.1 $\mathbf{R}_B$: Translational Anchor

The centre-of-mass position $\mathbf{R}_B \in \mathbb{R}^3$ is the bead's spatial location in the global (lab) frame. It is computed at mapping time as

$$\mathbf{R}_B = \frac{1}{M_B} \sum_{i \in S_B} m_i\, \mathbf{r}_i \tag{36}$$

and thereafter evolves under the equations of motion.

Physical role. $\mathbf{R}_B$ is the translational anchor from which every pairwise interaction begins. It enters:

- separation vectors: $\mathbf{r}_{AB} = \mathbf{R}_B - \mathbf{R}_A$, $r = |\mathbf{r}_{AB}|$
- neighbour-list construction and periodic image selection
- translational kinetic energy: $T_{\text{trans}} = \frac{1}{2} M_B \dot{\mathbf{R}}_B^2$
- the atomistic→CG mapping as the image of the fragment centroid

Update frequency. $\mathbf{R}_B$ is the most rapidly changing state variable. It is updated *every dynamics step* by the integrator.

Ownership. $\mathbf{R}_B$ belongs to the *kinematic shell* of the bead—the layer concerned with motion through space, not with intrinsic identity.

14.2 $\mathbf{Q}_B$: Orientation Frame

The orientation $\mathbf{Q}_B \in \mathrm{SO}(3)$ is a 3×3 orthonormal matrix whose columns are the bead's principal axes, ordered by ascending eigenvalue of the inertia tensor. It satisfies the invariant

$$\mathbf{Q}_B^T \mathbf{Q}_B = \mathbf{I} \tag{37}$$

which must be enforced (e.g. via Gram–Schmidt or quaternion renormalisation) at every mutation site.

Physical role. $\mathbf{Q}_B$ is the *pose operator* for the descriptor. The coefficients $\{c_{\ell m}^{(k)}\}$ are defined in the bead's local frame; $\mathbf{Q}_B$ tells the interaction engine how to rotate those coefficients into the frame of an interaction partner. Without $\mathbf{Q}_B$, the descriptor has no spatial meaning.

Concretely, $\mathbf{Q}_B$ enters:

- the relative rotation $R = \mathbf{Q}_A^T \mathbf{Q}_B$ used to transform B's coefficients into A's local frame
- torque evaluation: $\boldsymbol{\tau}_B = -\partial U / \partial \Omega_B$
- rotational equations of motion (Euler or quaternion propagation)
- the canonical orientation established during atomistic→CG mapping

Update frequency. $\mathbf{Q}_B$ is updated *every dynamics step*, in parallel with $\mathbf{R}_B$. After coarse-graining, the bead is an independent rigid anisotropic object; its orientation evolves dynamically and must not be recomputed from atomistic positions during a running simulation.

Ownership. $\mathbf{Q}_B$ belongs to the *kinematic shell*, alongside $\mathbf{R}_B$. Together they define the bead's pose—where it is and how it is oriented.

Degeneracy considerations. For highly symmetric or near-linear fragments the inertia tensor may have degenerate eigenvalues, making the principal-axis assignment ambiguous. The implementation must detect this case (eigenvalue ratio $\lambda_i/\lambda_j < \epsilon_{\mathrm{degen}}$) and fall back to a convention-based axis assignment to ensure reproducibility.

14.3 $\{c_{\ell m}^{(k)}\}$: Descriptor Coefficients

The spherical harmonic coefficients are the bead's *intrinsic identity*. Everything else in the state is either motion or bookkeeping; the coefficients encode what the bead actually is.

Physical role. For each channel $k \in \{\text{steric, electrostatic, dispersion}\}$ the surface field is expanded as

$$S_B^{(k)}(\theta, \phi) = \sum_{\ell=0}^{\ell_{\max}} \sum_{m=-\ell}^{\ell} c_{\ell m}^{(k)} Y_{\ell m}(\theta, \phi) \tag{38}$$

The coefficients carry physically distinct information:

- **Steric:** geometric accessibility and excluded-volume pattern
- **Electrostatic:** anisotropic charge-field structure
- **Dispersion:** anisotropic attractive strength distribution

Internal structure. The coefficient set is not a flat list. It has a channel-indexed, band-structured hierarchy:

$$\{c_{\ell m}^{(k)}\} \implies \underbrace{k}_{\text{channel}} \to \underbrace{\ell}_{\text{angular order}} \to \underbrace{m}_{\text{azimuthal mode}} \tag{39}$$

This nesting is not cosmetic. Adaptive truncation, channel activation, and per-ℓ kernel evaluation all operate naturally on this structure.

Update frequency. For rigid fragments the coefficients are *generated once* during atomistic→CG mapping and remain fixed thereafter. They change only when:

- the adaptive truncation loop promotes $\ell_{\max}$
- the underlying fragment changes (bond breaking, conformational rearrangement)
- the system is remapped

In a standard rigid-bead simulation, the coefficients are *static intrinsic data*, reused many times per dynamics step during pairwise interaction evaluation.

Ownership. $\{c_{\ell m}^{(k)}\}$ belongs to the *identity layer*—the layer that defines what the bead intrinsically is, independent of where it is or how it is currently oriented.

14.4 M_B: Total Mass

$$M_B = \sum_{i \in S_B} m_i \tag{40}$$

Physical role. M_B is the translational inertia scale. It enters:

- Newton's second law: $M_B \ddot{\mathbf{R}}_B = \mathbf{F}_B$
- translational kinetic energy: $T_{\text{trans}} = \frac{1}{2} M_B \dot{\mathbf{R}}_B^2$
- momentum conservation: $\mathbf{p}_B = M_B \dot{\mathbf{R}}_B$
- thermostat coupling (Langevin friction, Nosé–Hoover mass ratios)

Update frequency. M_B is a *conserved scalar* that changes only if the fragment definition changes. It must never drift during dynamics.

Ownership. M_B belongs to the *identity layer*, alongside the coefficients. It is not part of the descriptor machinery—it describes how the bead responds to force, not how it interacts anisotropically.

Separation principle. Descriptor coefficients tell you *how the bead interacts anisotropically.* Mass tells you *how the bead responds dynamically to force.* These are distinct responsibilities and must not be conflated.

14.5 Q_B^{tot}: Total Charge

$$Q_B^{\text{tot}} = \sum_{i \in S_B} q_i \tag{41}$$

Physical role. Q_B^{tot} is the net scalar charge of the bead. It enters:

- global charge-balance verification
- monopole contributions to long-range electrostatic treatments (Ewald summation, reaction-field corrections)
- system-level diagnostics and conservation checks

Relationship to the electrostatic channel. Q_B^{tot} and the electrostatic descriptor coefficients $\{c_{\ell m}^{(\text{elec})}\}$ are related but not identical:

- Q_B^{tot} is the *net charge*—a single scalar
- $c_{00}^{(\text{elec})}$ is proportional to the monopole moment of the electrostatic surface field
- higher-order coefficients $c_{\ell m}^{(\text{elec})}$ with $\ell \geq 1$ encode the *angular structure* of the electrostatic field

Both must be retained as explicit state variables. The scalar charge serves conservation and long-range bookkeeping; the coefficients serve anisotropic short-range interaction evaluation. The channel coefficients must not "replace" the total charge as an explicit bead property.

Update frequency. Q_B^{tot} is a *conserved scalar* that changes only if the fragment is remapped.

Ownership. Q_B^{tot} belongs to the *identity layer*.

14.6 State Layer Architecture

The preceding analysis reveals a natural four-layer decomposition of the bead state.

14.6.1 Layer A: Identity

The intrinsic properties of the bead, generated at mapping time and thereafter fixed unless remapping occurs:

$$\mathcal{I}_B = \left\{ \{c_{\ell m}^{(k)}\},\ M_B,\ Q_B^{\text{tot}} \right\} \tag{42}$$

14.6.2 Layer B: Pose

The current spatial configuration of the bead, evolving under the equations of motion at every dynamics step:

$$\mathcal{P}_B = \left\{ \mathbf{R}_B,\ \mathbf{Q}_B \right\} \tag{43}$$

14.6.3 Layer C: Resolution

Metadata governing the trustworthiness and complexity of the descriptor. Not part of the mathematical state tuple, but logically attached to the coefficient block:

- active channel set: which of the three channels carry nonzero coefficients
- current $\ell_{\max}$ per channel
- reconstruction residual $\varepsilon_B^{(k)}$ per channel
- frame construction method and well-definedness flag

These determine whether the bead's descriptor is trustworthy at its current resolution level.

14.6.4 Layer D: Provenance

Debugging and traceability metadata, also not part of the mathematical state, but essential for an anti-black-box architecture:

- source fragment identifier (which atomistic fragment generated this bead)
- frame method used (inertia, bond-based, fallback)
- whether the frame was well-defined or degenerate
- the residual achieved at the time of descriptor generation
- original atom count and bond count

Provenance does not change the physics, but it determines whether the architecture is debuggable.

14.6.5 Layer summary

Layer	Contents	Update frequency	Role
A (Identity)	$\{c_{\ell m}^{(k)}\}$, M_B, Q_B^{tot}	Fixed unless remapped	What the bead is
B (Pose)	$\mathbf{R}_B$, $\mathbf{Q}_B$	Every dynamics step	Where and how it is orie
C (Resolution)	$\ell_{\max}$, active channels, residuals	At adaptation events	Whether the descriptor is
D (Provenance)	Fragment ID, frame method, diagnostics	At mapping time only	Traceability and debuggi

14.7 Update Frequency Classification

The four-layer decomposition induces a strict update-frequency hierarchy that prevents unnecessary recomputation.

Frequently updated (every step).

$$\mathbf{R}_B,\ \mathbf{Q}_B \tag{44}$$

These are the evolving kinematic variables. They change under force and torque at every integration step.

Infrequently updated (at adaptation events).

$$\{c_{\ell m}^{(k)}\},\ \ell_{\max},\ \varepsilon_B^{(k)} \tag{45}$$

For rigid fragments the coefficients remain fixed during normal dynamics. They change only when the adaptive truncation loop promotes $\ell_{\max}$ or the underlying fragment is modified.

Essentially fixed (conserved unless remapped).

$$M_B,\ Q_B^{\text{tot}} \tag{46}$$

These must not drift during dynamics. A change in either implies that the underlying atomistic fragment has changed.

This classification prevents the antipattern of recomputing every piece of bead state at every step—a design characteristic of systems built by implementations that are afraid of structure.

14.8 Interaction Engine View

When evaluating the pairwise energy U_{AB}, the interaction engine should receive a *projected physical view* of each bead, not the entire bead object indiscriminately. The engine consumes:

- **Relative position** from $\mathbf{R}_A$, $\mathbf{R}_B$:

$$\mathbf{r}_{AB} = \mathbf{R}_B - \mathbf{R}_A, \quad r = |\mathbf{r}_{AB}| \tag{47}$$

- **Orientation relation** from $\mathbf{Q}_A$, $\mathbf{Q}_B$:

$$R = \mathbf{Q}_A^T \mathbf{Q}_B \tag{48}$$

- **Descriptor content** from $\{c_{\ell m}^{(k)}\}_A$, $\{c_{\ell m}^{(k)}\}_B$

The engine does *not* need M_B or Q_B^{tot} directly, except where a separate monopole term or long-range electrostatic treatment requires the scalar charge.

This separation ensures that the interaction engine operates on the physically relevant projection of the bead state, not on an undifferentiated data dump.

14.9 Dynamics Engine View

The motion engine consumes a different projection of the bead state:

- $\mathbf{R}_B$: position to be integrated
- $\mathbf{Q}_B$: orientation to be integrated
- M_B: translational inertia for force→acceleration conversion
- $\mathbf{I}_B$: rotational inertia for torque→angular acceleration conversion (see below)
- $\mathbf{F}_B$, $\boldsymbol{\tau}_B$: force and torque from the interaction engine

The dynamics engine does *not* need to know about descriptor generation details, channel sampling procedures, or spherical harmonic projection methodology. It needs the resulting force and torque, not a reconstruction of how the steric channel was projected onto harmonics.

This separation is healthy: the interaction engine converts descriptor content into forces and torques; the dynamics engine converts forces and torques into motion.

14.10 Rotational Inertia Extension

The formal state as written is sufficient for descriptor representation and interaction evaluation. However, for complete rigid-body rotational dynamics the bead requires one additional quantity: the *rotational inertia tensor*.

For a rigid bead composed of atoms $i \in S_B$ with positions $\boldsymbol{\delta}_i = \mathbf{r}_i - \mathbf{R}_B$ relative to the centre of mass:

$$\mathbf{I}_B = \sum_{i \in S_B} m_i \left(|\boldsymbol{\delta}_i|^2 \, \mathbf{I}_3 - \boldsymbol{\delta}_i \boldsymbol{\delta}_i^T \right) \tag{49}$$

Diagonalisation in the principal-axis frame yields:

$$\mathbf{I}_B = \mathbf{Q}_B \, \mathrm{diag}(I_1, I_2, I_3) \, \mathbf{Q}_B^T \tag{50}$$

The principal moments (I_1, I_2, I_3) are scalars that, like M_B, are generated at mapping time and remain fixed unless the fragment changes.

With this extension, Euler's rotational equations of motion become:

$$I_\alpha \dot{\omega}_\alpha = \tau_\alpha^{\mathrm{body}} - \epsilon_{\alpha\beta\gamma} \, \omega_\beta \, I_\gamma \, \omega_\gamma \qquad (\alpha = 1, 2, 3) \tag{51}$$

where ω_α are the body-frame angular velocity components and $\tau_\alpha^{\mathrm{body}}$ are the body-frame torque components.

The theoretically complete dynamic bead state is therefore:

$$\boxed{B = \left\{ \mathbf{R}_B, \ \mathbf{Q}_B, \ \{c_{\ell m}^{(k)}\}, \ M_B, \ Q_B^{\mathrm{tot}}, \ \mathbf{I}_B \right\}} \tag{52}$$

This is the state of a genuinely physical anisotropic bead—not merely a clever interaction shell, but an object with full translational and rotational dynamics grounded in the inertia of its underlying atomistic fragment.

14.11 Conceptual Summary

The bead state admits a clean three-way separation:

$$\underbrace{\mathbf{R}_B, \ \mathbf{Q}_B}_{\text{where and how it is oriented}} \quad \Big| \quad \underbrace{\{c_{\ell m}^{(k)}\}}_{\text{what it intrinsically is}} \quad \Big| \quad \underbrace{M_B, \ Q_B^{\mathrm{tot}}, \ \mathbf{I}_B}_{\text{how it responds to force and torque}} \tag{53}$$

This separation must not be blurred:

- The **pose** $(\mathbf{R}_B, \mathbf{Q}_B)$ is dynamic—it evolves every step.
- The **descriptor** $(\{c_{\ell m}^{(k)}\})$ is intrinsic—it encodes the bead's angular identity and changes only under remapping or adaptation.
- The **inertial properties** $(M_B, Q_B^{\mathrm{tot}}, \mathbf{I}_B)$ are conserved scalars inherited from the atomistic fragment.

An effective implementation treats these three groups with distinct storage strategies, distinct access patterns, and distinct update schedules. The interaction engine sees pose and descriptor. The dynamics engine sees pose and inertia. Neither needs to see the other's internal details. That is the right architecture.

15 Conclusion

This document has developed the anisotropic surface-mapped bead from initial concept to a formally specified, architecturally decomposed physical object.

The progression is intentional and cumulative:

1. **Atomistic-to-CG mapping** established the centre-of-mass, local frame, and aggregate scalars that anchor every bead to its underlying fragment.
2. **Surface descriptor representation** replaced isotropic point particles with directional surface fields, expanded in spherical harmonics for compact storage and efficient evaluation.
3. **Orientation-coupled interactions** introduced relative alignment dependence, torque, and rotational dynamics—turning the bead from a passive site into a dynamical object with angular degrees of freedom.
4. **Multi-channel descriptor enrichment** decomposed the surface field into steric, electrostatic, and dispersion channels, enabling physically distinct interaction mechanisms to be resolved independently.
5. **Unified descriptor strategy** eliminated discrete model switching in favour of a single continuous formalism with residual-driven adaptive truncation.
6. **Atomistic preparation layer** formalised the scale boundary via the fragment view, ensuring that every descriptor coefficient is traceable to specific atoms and that invalid inputs are rejected with explicit status codes.
7. **Implementation specification** fixed the per-ℓ channel kernels, the Wigner-D rotation pipeline, and the adaptive refinement loop into a rigid, deterministic architecture.
8. **Formal bead state decomposition** analysed each variable in the canonical tuple, established the four-layer architecture (identity, pose, resolution, provenance), classified update frequencies, defined the interaction and dynamics engine views, and extended the state to include rotational inertia for complete rigid-body dynamics.

The result is a bead defined by the complete state

$$B = \left\{\mathbf{R}_B,\ \mathbf{Q}_B,\ \{c_{\ell m}^{(k)}\},\ M_B,\ Q_B^{\mathrm{tot}},\ \mathbf{I}_B\right\} \tag{54}$$

with clear ownership, update rules, and numerical roles for every term.

15.1 Scope Boundary

This document deliberately stops at the *isolated bead*—its internal structure, its descriptor, its interactions with other beads, and its rigid-body dynamics. The bead as defined here is a self-contained physical object: it knows what it is, how it is oriented, how it interacts, and how it moves.

What the bead does *not* yet know is its environment. It has no awareness of local density, coordination, orientational order, or collective state. It cannot distinguish between being in a gas, a liquid, a crystal, or at an interface. It cannot respond to crowding, stiffen under compression, or relax into alignment with its neighbours.

These capabilities require a second layer of state—environment-responsive observables and slow internal variables—that couples the bead to its local surroundings. This extension is the subject of a companion document:

Environment-Responsive Coarse-Grained Bead Dynamics
(VSEPR-SIM Project, `section_environment_responsive_beads.tex`)

That document introduces the extended bead state $\mathbb{B}_B = \{B,\, X_B\}$, where X_B contains environment observables (local density, coordination, orientational order) and a slow internal state variable η_B with memory and relaxation dynamics. The coupling of X_B back into the interaction kernels enables density-dependent behaviour, spontaneous ordering, hysteresis, and metastability—the transition from "the model works" to "the system behaves."